

PRAGMATIC REWARDS: INTRINSIC COMMUNICATIVE CLARITY IN MULTI-AGENT REINFORCEMENT LEARNING

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ABSTRACT

Reinforcement Learning (RL) agents often fail in cooperative settings because their actions, while individually optimal, are unpredictable to their partners. This challenge is magnified in human-agent teams, where an agent’s opacity can erode trust and team effectiveness. We address this limitation by establishing a formal equivalence between pragmatic inference, as formalized by the Rational Speech Act (RSA) framework, and Maximum Entropy Reinforcement Learning (Max-Ent RL). This theoretical bridge yields a principled intrinsic reward for “legible” behavior—actions that clearly communicate an agent’s private state—computed offline by a Pragmatic Reward Module (PRM) that adds no latency to decision-making. The framework exposes the literal listener L_0 , the assumed common ground against which clarity is measured, as its central design choice. We study a spectrum of listeners, from hand-crafted geometric heuristics to a listener learned jointly with the policy, and show that the learned instantiation turns the pragmatic reward into a variational lower bound on the mutual information between an agent’s intention and its observable behavior, unifying pragmatic and information-theoretic approaches to implicit communication. We evaluate on two environments separated by an oracle-gap criterion that bounds the task value of communication: a cooperative-navigation control where the gap is near zero, and scout-support, a new tractable benchmark modeled on pickup-and-delivery teams in which a scout’s private target only pays off if a slower teammate infers it early. At converged training budgets the results reverse the intuition that better listeners make better rewards: simple motion-grounded listeners close 60–70% of the oracle premium through the speaker-side reward alone—their crude, overtly geometric notion of clarity is exactly what a teammate can read from raw observations—while a learned listener becomes too competent, decoding any behavior at 97%, saturating its reward, and exerting no pressure on the policy; deploying it instead as the teammate’s “ear” at execution recovers a third of the premium. We distill this into an audience-matching principle anticipated by the structure of RSA itself: the literal listener anchoring a legibility

reward must be bounded—no stronger than the weakest intended reader of the behavior. We further document how pre-convergence evaluation produced multiple spurious reversals of these conclusions, a hazard specific to co-adaptive communication measures. Our framework provides a formal model, a benchmark, and a design principle for agents that are not only effective but also strategically collaborative.

Index Terms— Reinforcement Learning, Intrinsic Motivation, Multi-Agent Systems, Computational Pragmatics, Legibility

1. INTRODUCTION

Human collaboration is fundamentally a goal-oriented, strategic activity. To succeed, we do not just act to achieve our own goals; we act in ways that allow our partners to infer our beliefs and intentions, enabling better coordination. The same is true for artificial agents. In cooperative multi-agent reinforcement learning (MARL) environments, an agent’s success depends not only on its own actions but also on being understood by its teammates. Consider a cooperative navigation task in which each agent must cover a target of a preferred type: an agent that simply rushes toward its target may take a path that is ambiguous from a teammate’s perspective, causing the teammate to abandon a target the agent never intended to claim. A more sophisticated agent might take a slightly suboptimal path that unambiguously singles out its intended target, making its intention clear and enabling better team coordination.

This property of being easily understood is called “legibility” [1]. However, standard RL agents, optimized solely for an extrinsic task reward, have no incentive to be legible. This leads to our central question: how can we formally encourage an RL agent to choose actions that are not just extrinsically optimal for the task, but also intrinsically optimal for communicating its private state or intentions?

This work proposes a solution in the form of an intrinsic reward, R_{comm} , that quantifies the communicative clarity of an agent’s actions. To calculate this re-

ward, we model an “observer” (e.g., another agent) who uses pragmatic inference to reason about why the agent chose a particular action over the available alternatives. We find the ideal formalism for this observer in the Rational Speech Act (RSA) framework [2], which provides a formal account of reasoning about communicative intent.

Any such observer model must be anchored in a literal listener L_0 : the assumed common ground that specifies how behavior would be interpreted in the absence of pragmatic reasoning. Our analysis and experiments show that this choice is not an implementation detail but the decisive factor in whether pragmatic rewards help or hurt. Hand-crafted listeners can be systematically misaligned with what the policy can express, incentivizing costly signaling that an actual observer cannot decode. This motivates our second contribution: replacing the hand-crafted L_0 with a listener learned from the agents’ own behavior, which guarantees, by construction, that rewarded actions are decodable.

Our contributions are as follows:

- We establish a formal equivalence between the objective functions of Max-Ent RL and the RSA framework, allowing the RSA “listener value” to be imported as a principled intrinsic reward for legibility, and we show that purely communicative models such as CRSA [3] are recovered as a myopic special case ($\gamma = 0$, $R_{\text{ext}} = 0$).
- We operationalize the theory as a Pragmatic Reward Module (PRM): an offline component, compatible with centralized training and decentralized execution (CTDE), that computes R_{comm} from replayed transitions and adds no latency at execution time.
- We characterize the space of literal listeners and prove that instantiating L_0 with a listener trained by maximum likelihood on the agents’ behavior makes the expected pragmatic reward a variational lower bound on the mutual information $I(M; S, A)$ between private intentions and observable behavior. This connects pragmatic rewards to information-regularization [4] and skill-discriminability [5] objectives, and positions them as the literal (0-th order) special case of our framework.
- We propose an oracle-gap criterion for benchmarking communicative rewards—the gap between a baseline and an oracle given all private states bounds the task value of communication—and introduce scout-support, a tractable two-agent benchmark with a large, structural gap: a scout’s private target only pays off if a slower supporter infers it early, while a pickup waypoint makes

the scout’s task-optimal motion uninformative exactly when information is needed. A cooperative-navigation control with near-zero gap completes the pair.

- Empirically, we dissect implicit communication into speaker and receiver contributions at converged training budgets. Motion-grounded hand-crafted listeners (progress, filter, plain cosine) close 60–70% of the oracle premium through the speaker-side reward alone (Welch $t \geq 5$ at 10 seeds), with the largest gains in the teammate’s commitment timing; a learned listener saturates (97% decoding of any behavior) and shapes nothing, while its deployment as the teammate’s execution-time ear recovers a significant 30%. We distill this into an audience-matching principle: the listener anchoring a legibility reward must be no stronger than the intended reader—boundedness, the defining property of RSA’s literal listener, is what makes the pragmatic gradient informative. We also document, as a methodological hazard, that pre-convergence evaluation of these co-adaptive systems produced repeated spurious reversals.

2. RELATED WORK

Our framework, which uses pragmatic inference to generate an intrinsic reward for communicative clarity, builds on four lines of research: (1) legibility and observer-aware planning; (2) intrinsic rewards for multi-agent coordination; (3) implicit and emergent communication; and (4) computational pragmatics.

2.1. Legibility, Goal Recognition, and Observer-Aware Planning

Inferring an agent’s latent state (e.g., its goal) from observed behavior is the central problem of Inverse Reinforcement Learning (IRL) [6] and computational Theory of Mind (ToM) [7, 8]. Legibility inverts this problem: the actor shapes its own behavior so that an observer’s inference succeeds. Originally formalized for robot motion [1], legibility has been extended to stochastic sequential decision problems [9], unified with explicability and predictability in observer-aware MDPs [10, 11], and recently brought to MARL through legible variants of interactive POMDPs [12]. These approaches typically embed the observer model in the planning objective, requiring inference over goals at decision time. In contrast, our framework converts legibility into an intrinsic reward computed offline during training, so the executed policy carries its communicative competence “for free,” with no observer model, belief tracking, or additional computation in the execution loop.

A complementary line of work uses ToM as a source of intrinsic motivation [13], rewarding agents for accurately predicting the beliefs of others; this is listener-centric, rewarding agents for being predictive of others. Our pragmatic framework is speaker-centric: it rewards agents for making their own private state clear, regardless of their partners’ modeling ability.

2.2. Intrinsic Rewards for Multi-Agent Coordination

Our pragmatic reward is a form of intrinsic motivation for cooperation, and is thus related to social influence [14], which rewards an agent for causally changing its partners’ behavior. The communicative goals differ: influence rewards an agent for compelling a reaction, whereas the pragmatic reward incentivizes the successful transmission of information about oneself, decoupling the speaker’s reward from the listener’s reaction. Teammates remain free to exploit or ignore the revealed intention, which promotes a more flexible form of coordination.

Closest to our learned-listener instantiation is the information-regularization framework of Strouse et al. [4], which adds a mutual-information term $I(A; G|S)$ to the objective so that agents share (or hide) their goal G through their actions. Similarly, Eccles et al. [15] add a positive-signaling bias encouraging messages to depend informatively on observations, and, in the single-agent setting, DIAYN [5] rewards skills for being discriminable from states via a learned classifier. Proposition 1 makes the relationship precise: these MI-based objectives correspond to the literal listener level of our framework, obtained when the RSA recursion depth is zero and L_0 is trained by maximum likelihood. The pragmatic levels add contrastive reasoning over the alternative actions the agent could have taken—a model of communication richer than statistical dependence alone. Our framework also shares the goal of grounded, convention-free coordination with Bayesian action decoding [16] and off-belief learning [18], which control how much agents may read into their partners’ actions; we instead directly reward the actor for making the intended reading easy.

2.3. Implicit Communication vs. Explicit Channels

A significant body of MARL research equips agents with a dedicated communication channel and learns an emergent protocol end-to-end [19, 20, 21]. Our framework diverges from this paradigm by focusing on implicit communication: there is no separate channel and no explicit message passing. An agent “speaks” by moving in the environment, and a teammate “listens” by observing that movement. The pragmatic reward incentivizes actions that are simultaneously effective for

the task and clear as signals of intent. This is valuable when explicit communication is costly, unavailable, or must be fused with physical action, as is common in human-agent teams. Because the signal is the action, the exploration-communication conflict identified in emergent-communication work [15, 17] appears here as a reward-shaping trade-off, which we analyze through the weight λ .

2.4. Computational Pragmatics

RSA [2] formalizes communication as recursive probabilistic reasoning between a speaker and a listener, and has been given a rate-distortion interpretation [22]. Neural instantiations replace the hand-specified literal listener with learned models [23, 24], a design choice our learned- L_θ instantiation transposes to the behavioral domain. The Collaborative RSA (CRSA) model [3] extends RSA to multi-turn dialog between agents with asymmetric private information. We show in Section 4.6 that such purely communicative models arise as the myopic, task-free special case of our strategic framework, which extends pragmatic reasoning to long-horizon sequential decision-making where instrumental and communicative goals are intertwined.

3. FORMALISMS FOR ACTION AND INTERPRETATION

To model legible strategic behavior we require two formalisms: one for optimal sequential decision-making under uncertainty, and one for how an observer infers intent from an action. For the former we use Max-Ent RL; for the latter, RSA.

3.1. Maximum Entropy RL

Max-Ent RL extends the standard RL objective with the intuition “be as versatile as possible while still being successful.” In a single-step problem, the agent seeks a policy $\pi(a|s)$ maximizing expected reward plus policy entropy [25]:

$$J(\pi) = \mathbb{E}_{a \sim \pi(\cdot|s)}[R(s, a)] + \alpha_{\text{RL}} H(\pi(\cdot|s)). \quad (1)$$

The optimal policy is a softmax over rewards [26], with temperature α_{RL} trading off exploitation against versatility:

$$\pi^*(a|s) \propto \exp(R(s, a)/\alpha_{\text{RL}}). \quad (2)$$

3.2. The Rational Speech Act Framework

RSA [2] models an observer who infers an agent’s intended meaning not from the literal content of an action alone, but by reasoning about why the agent chose that

action over the alternatives: “if they had intended X , they would have done Y ; since they did Z , they must intend something else.” Formally, RSA seeks a communicative policy $S(u|m)$ that optimally balances communicative success against action cost, yielding a softmax over the utility, or “listener value,” of each communicative act:

$$S(u|m) \propto \exp(\alpha_{\text{RSA}} \cdot V_L(u, m)), \quad (3)$$

where $V_L(u, m)$ quantifies the clarity of act u for conveying meaning m to an observer, penalized by the act’s cost.

4. REWARDING LEGIBILITY

Our agent optimizes a composite reward balancing task objectives with communicative clarity. This section defines the reward via a formal bridge between pragmatic inference and RL, characterizes the space of listener models that anchor it, describes the architecture that computes it, and presents the resulting strategic Bellman equation.

4.1. The Total Reward

At each timestep t the agent receives the standard extrinsic task reward $R_{\text{ext}}(s_t, a_t)$ from the environment, and generates internally an intrinsic communicative reward $R_{\text{comm}}(s_t, a_t)$ quantifying how clearly action a_t communicates the agent’s private state m_t (its “meaning,” e.g., an intended target) to an imagined observer. The policy maximizes the weighted sum

$$R_{\text{total}}(s_t, a_t) = R_{\text{ext}}(s_t, a_t) + \lambda R_{\text{comm}}(s_t, a_t), \quad (4)$$

where λ balances task success against communicative clarity. The central question is how to define R_{comm} in a principled way.

4.2. Deriving R_{comm} via Pragmatic Inference

We treat the agent’s action a as a communicative “utterance” and its private state m as the “meaning” to be conveyed (Table 1). At first glance, the objective of an RL agent and the goal of a pragmatic speaker seem unrelated. However, the optimal single-step Max-Ent policy (Eq. 2) and the RSA speaker (Eq. 3) are both softmax distributions, identical under the re-indexing $\alpha_{\text{RSA}} = 1/\alpha_{\text{RL}}$. This formal alignment is the cornerstone of our framework: it justifies importing the well-defined listener value V_L from computational pragmatics as a principled intrinsic reward. We define

$$R_{\text{comm}}(s_t, a_t) \triangleq V_L(a_t, m_t) = \log L_t^*(m_t|a_t, s_t) - C(a_t), \quad (5)$$

Table 1. Mapping between Max-Ent RL and RSA components.

Max-Ent RL component	RSA analogue
State (s)	Context
Action (a)	Utterance (u)
Private state / intention (m)	Meaning (m)
Agent policy ($\pi(a s, m)$)	Speaker ($S(u m)$)
Intrinsic reward (R_{comm})	Listener value (V_L)
Temperature (α_{RL})	Rationality ($1/\alpha_{\text{RSA}}$)

where $L_t^*(m_t|a_t, s_t)$ is the probability a pragmatic listener assigns to the agent’s true private state after observing its action in context s_t , and $C(a_t)$ is an optional action cost. In practice we set $C(a_t) = 0$: the cost of a legible action is already handled by the task reward, since a less direct but clearer path is naturally penalized by a lower R_{ext} . In our implementation we also add the constant $\log |\mathcal{M}|$, so that R_{comm} reads as the information gain over a uniform prior: it is zero when the action is uninformative and positive when the listener’s posterior concentrates on the true meaning.

4.3. The Space of Literal Listeners

Computing L^* requires anchoring the pragmatic recursion in a literal listener $L_0(m|a, s)$: the assumed common ground specifying how behavior would be interpreted absent pragmatic reasoning. Given L_0 and a set $\mathcal{A}$ of alternative actions (the true action plus K sampled alternatives), the pragmatic listener is obtained by N_r steps of RSA alternation,

$$S_k(a|m) = \frac{L_{k-1}(m|a)}{\sum_{a' \in \mathcal{A}} L_{k-1}(m|a')}, \quad L_k(m|a) = \frac{S_k(a|m)}{\sum_{m'} S_k(a|m')}, \quad (6)$$

and $L^* = L_{N_r}$. The choice of L_0 is the framework’s most consequential free parameter, and we study three instantiations.

(i) Direction listeners. A computationally cheap heuristic interprets actions by their geometric similarity to ideal paths. For agent position p and each meaning m , let v_m be the unit vector toward the nearest target consistent with m . A naïve score is $S(a, m) = \alpha_{\text{temp}} \cos(a, v_m)$ (simple). To capture unambiguity explicitly, an exclusivity variant penalizes similarity to competing meanings,

$$S_{\text{excl}}(a, m) = \alpha_{\text{temp}} (\cos(a, v_m) - \max_{m' \neq m} \cos(a, v_{m'})), \quad (7)$$

and $L_0(\cdot|a) = \text{softmax}_m S_{\text{excl}}(a, \cdot)$. This listener rewards actions aligned with the ideal path toward the true target and misaligned with the ideal paths of all confusable targets.

(ii) Progress listener. Instead of instantaneous direction, this listener scores the progress an action realizes toward each candidate target: $S_{\text{prog}}(a, m) \propto d_{t-1}(m) - d_t(m)$, the per-step reduction in distance to the target consistent with m , normalized by the maximum step length. This mirrors the cost-to-go formulation of legibility in robot motion [1], which scores partial trajectories by how efficiently they advance each candidate goal.

(iii) Filter listener. Instantaneous listeners ignore history: an action that is uninformative in isolation may complete a legible trajectory. The filter listener maintains a recursive belief over meanings, updated multiplicatively from per-step directional likelihoods with exponential forgetting,

$$b_t(m) \propto b_{t-1}(m)^\rho \exp(\alpha_{\text{temp}} \cos(a_t, v_m)), \quad (8)$$

and rewards the log-belief in the true meaning, $R_{\text{comm}} = \log b_t(m_t) + \log |\mathcal{M}|$. Under discounting this reproduces the trajectory-integral character of legibility—early clarity is worth more than late clarity—and makes R_{comm} explicitly non-Markovian, which the PRM accommodates because rewards are computed offline.

(iv) Learned listener. Hand-crafted listeners presuppose that the features they score are the ones an actual observer would use. To remove this assumption, we train a network $L_\theta(m|s, a)$ by maximum likelihood on the replay buffer to predict an agent’s true meaning from its publicly observable context and action (the private state is masked from the input). The listener is trained concurrently with the policy, so the common ground co-adapts: the reward only credits signals that are actually decodable from the agents’ current behavioral distribution. With $N_r = 0$ the reward is simply $R_{\text{comm}} = \log L_\theta(m_t|s_t, a_t) + \log |\mathcal{M}|$; with $N_r \geq 1$ the RSA recursion (Eq. 6) is applied on top of L_θ , adding contrastive reasoning over alternatives.

The learned instantiation admits an information-theoretic characterization.

Proposition 1. Let meanings be drawn uniformly on $\mathcal{M}$, and let $L_\theta(m|s, a)$ be any conditional distribution. Then, under the joint distribution induced by the policy,

$$\mathbb{E}[\log L_\theta(m|s_t, a_t) + \log |\mathcal{M}|] \leq I(M; S_t, A_t), \quad (9)$$

with equality iff $L_\theta(m|s, a) = p(m|s_t = s, a_t = a)$. Hence the expected learned-listener reward is the Barber–Agakov variational lower bound [27] on the mutual information between the agent’s private meaning and its observable behavior; maximum-likelihood training of θ tightens the bound, and policy optimization against it maximizes information disclosure.

Proof. $\mathbb{E}[\log L_\theta(m|s, a)] = \mathbb{E}[\log p(m|s, a)] - \mathbb{E}_{s,a}[\text{KL}(p(\cdot|s, a) \| L_\theta(\cdot|s, a))]$. Adding $H(M) = \log |\mathcal{M}|$ gives the

claim; the KL term vanishes iff L_θ matches the true posterior. $\square$

Proposition 1 unifies two families of methods. The cooperative information regularizer of Strouse et al. [4], the positive-signaling bias of Eccles et al. [15], and the skill discriminator of DIAYN [5] all maximize variational MI bounds of exactly this form; within our framework they are the literal ($N_r = 0$) level of pragmatic reasoning with a learned common ground. Conversely, the RSA recursion is a strict generalization that scores an action not by how much information it carries in isolation, but by how it contrasts with the alternatives the agent declined—the signature of pragmatic communication [2, 22].

4.4. The Pragmatic Reward Module

The theoretical bridge yields a practical hybrid architecture that decouples action selection from communicative analysis. The system comprises the main RL agent, which learns $\pi(a_t|s_t)$, and a Pragmatic Reward Module (PRM), which computes R_{comm} to shape the agent’s learning updates. The pragmatic analysis is performed offline during training—for static listeners, when a transition is collected; for the learned listener, when it is replayed, so the reward tracks the co-adapting common ground—and never in the action-selection loop, adding no latency to decision-making. We operate under CTDE: during centralized training the PRM has access to all agents’ private states, enabling a gold-standard legibility reward; at execution, agents act from local observations only, and the PRM is discarded.

4.5. The Strategic Bellman Equation

With the composite reward defined, the action-value function of the legible agent satisfies the soft Bellman equation

$$Q_t(s_t, a_t) = R_{\text{total}}(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}} \left[\alpha \log \sum_{a_{t+1}} \exp \left(\frac{1}{\alpha} Q_{t+1}(s_{t+1}, a_{t+1}) \right) \right], \quad (10)$$

which is the standard Max-Ent backup applied to R_{total} . The equation makes explicit the trade-off between immediate gains (task and clarity) and the long-term strategic value of future states: an agent may accept a lower immediate reward (e.g., a “corralling” detour) if the resulting state carries higher expected value through improved team coordination.

4.6. Generality: Recovering Communicative Agents

Communicative agents arise as a special case. Setting $R_{\text{ext}} = 0$ and $\lambda = 1$ makes the total reward purely

communicative; setting $\gamma = 0$ makes the agent myopic, collapsing Eq. 10 to the immediate reward plus policy entropy. This objective is exactly the per-turn gain function of the Collaborative RSA model [3], demonstrating that single-turn pragmatic models are a specific instance of our framework, which extends pragmatic reasoning to long-horizon sequential settings where instrumental and communicative goals interact.

5. EXPERIMENTAL SETUP

5.1. Environments

A meaningful benchmark for communicative rewards must certify that communication has task value at all. We therefore adopt an oracle-gap criterion: comparing a baseline against an oracle that receives all private states in its observations measures the maximum task reward that intent transmission can unlock. We evaluate on two environments, one on each side of this criterion.

Environment A: scout-support (oracle gap ≈ 0.5 per step). A two-agent team in a continuous 2D arena, modeled on scout-and-support patterns in search-and-rescue and warehouse logistics: a fast scout privately knows which of three sites is the true target, and a slower supporter earns the team a bonus only while both agents occupy the target together. Crucially, the scout must first visit a central pickup waypoint before its presence at the target counts (a forklift fetches the pallet before the dock). During this first leg, the scout’s task-optimal motion is identical regardless of target—structurally uninformative—so any early information the supporter receives must be actively signaled, e.g. by curving the approach to the waypoint toward the future destination, the anticipatory motion studied in robot legibility [1]. Sites lie on a circle (radius 1.0, 120° apart, jittered); the supporter’s lower speed (0.6 vs. 1.0) makes late commitment costly. The reward is the rendezvous bonus minus both agents’ distances to their current objectives; episodes last 50 steps. The supporter observes the scout’s position and velocity but never its target. Meanings are the three sites; only the scout receives R_{comm} .

Environment B: preference spread (oracle gap ≈ 0). Our earlier cooperative-navigation control: three agents and three colored landmarks, each agent holding a private color preference; the team is rewarded for matched coverage, penalized for distance and collisions. Because proximity resolves most coordination and acting on one’s own preference is extrinsically rewarded, an oracle gains nothing—communication here has no task value to unlock, which is precisely what makes it a useful control for measuring what a communicative reward costs.

5.2. Models and Conditions

All agents are trained with multi-agent Soft Actor-Critic (MASAC)—parameter-shared Gaussian actors (roles distinguished by a one-hot flag) and a centralized twin critic over joint observations and actions, with automatic entropy tuning—the natural multi-agent extension of the Max-Ent framework our theory is built on. Conditions differ only in reward and observation content: baseline ($\lambda = 0$); oracle ($\lambda = 0$, private states shared through observations); the four hand-crafted listeners of Section 4.3—simple, exclusivity (both with one RSA step over $K = 64$ sampled alternatives), progress, and filter; and the two learned ones, learned ($N_r = 0$) and learned + RSA (one recursion step over 16 alternatives). L_θ is a 2-layer MLP trained by cross-entropy on replayed transitions with the private state masked from its input. In Environment A we add ear conditions that append the listener’s posterior to the supporter’s observation at execution (Section 6.1), and a value-of-information weighting that scales R_{comm} by the supporter’s decision relevance (full weight before the scout’s pickup, down-weighted after). Environment B uses $\lambda = 0.1$ over 150 training cycles (480k steps) with the direction-based and learned listeners; Environment A uses $\lambda = 0.3$ over 400 cycles, with the headline conditions run at 10 seeds and the rest at 3. Full hyperparameters are in Table 4.

5.3. Evaluation Metrics

We evaluate periodically on 64 held-out episodes with deterministic policies. Task performance is the mean per-step team reward and its components. In Environment A, commitment accuracy is the fraction of steps in which the supporter’s nearest site is the true target—a direct behavioral readout of how early it decodes the scout. In Environment B, team specialization is the fraction of landmarks claimed uniquely (each agent assigned to its nearest landmark); a score of 1 indicates perfect division of labor. Legibility is measured by a fixed probe listener—the simple geometric listener, used identically across all conditions—reporting the accuracy with which an agent’s true preference is recovered from its instantaneous action (probe intent accuracy). The probe is computed over all steps, including those where an agent already rests on its target and its actions are uninformative, so absolute values understate legibility; because the probe is fixed and external to all training conditions, however, relative improvements reflect genuinely clearer behavior rather than adaptation to the metric. For learned-listener conditions we additionally report L_θ ’s own held-out accuracy, and for all PRM conditions the achieved communicative reward R_{comm} . Finally, we measure post-hoc decodability: for every trained run, a fresh listener network is trained from scratch on

deterministic rollouts of the final policy and evaluated on held-out episodes. This quantifies how much intent information each condition’s converged behavior carries under an identical decoding budget, independently of any listener used during training.

6. RESULTS

We present Environment A—where the oracle gap certifies that communication has task value—as the primary benchmark, and Environment B as the control that isolates what a communicative reward costs when there is nothing to say. Across both, the results separate cleanly along the common-ground axis: what matters is not whether a pragmatic reward is used, but which listener anchors it.

6.1. Environment A: When Communication Pays

Table 2 summarizes Environment A at the converged budget (400 cycles; 10 seeds for all key conditions); Figure 1 shows the time course of the supporter’s understanding and Figure 2 the information carried by the scout’s behavior. The converged oracle premium is 0.112 per step (0.957 vs. 0.845; commitment .880 vs. .851)—the irreducible value of early intent information once both teams have fully learned the task.

Motion-grounded hand-crafted listeners close most of the recoverable gap. The strongest conditions are the simplest: the progress listener closes 70% of the oracle premium ($0.924 \pm .008$, Welch $t = 6.2$ vs. baseline), the filter listener 67% ($t = 6.5$), and even the plain cosine listener 62% ($t = 5.0$)—through the speaker-side reward alone, with the largest commitment gains of any condition (.857–.861) and visibly more legible scouts (probe accuracy .61 vs. .54; Fig. 2). The mechanism is direct: these listeners only credit overtly geometric signals—motion biased toward the true target—and overt signals are exactly what the supporter’s own policy network can read from raw observations and motion history. No dedicated receiver machinery is needed; the channel is the behavior itself.

A too-competent listener exerts no pressure. The learned listener inverts this logic and fails by succeeding. Trained to convergence, L_θ decodes the scout at 97% whatever the scout does: its reward saturates near the log 3 ceiling (+1.03), the pragmatic gradient on behavior vanishes, and the speaker-only learned condition ends indistinguishable from—nominally below—the baseline (0.828, −15% closure, n.s.). Deploying the same listener as the supporter’s ear (its posterior appended to the supporter’s observation, still decentralized-executable) recovers a significant 30% of the premium ($0.878 \pm .012$, Welch $t = 2.15$), and the ear works nearly as well with

no speaker subsidy at all (0.874, 26%): with a saturated listener, all the value is in the decoding, none in the speaking. Combining the two families—the filter reward for the scout with its belief as the ear—matches the filter reward alone (0.920, 66%): once the speaker signals overtly, the equipped ear is redundant, confirming that the overt channel is read directly from behavior.

Robustness and grounding. The $\lambda \times \text{VOI}$ response surface is flat at convergence (0.83–0.88 for $\lambda \in [0.1, 0.6]$, all VOI weightings) until $\lambda = 1$ becomes destructive (0.61); the weighting effects that mattered mid-training (Sec. 6.1.1) wash out asymptotically. Zero-shot cross-play—pairing scouts and supporters from independently trained seeds—retains 92–94% of self-play reward in every condition, no worse than the baseline’s own cross-seed transfer; cross-paired filter teams (0.844) still match baseline self-play. The emergent protocols are thus grounded in task geometry rather than seed-specific convention, addressing the fragility concern raised by the zero-shot-coordination literature [18]. Finally, at $16\times$ the parameter count (hidden width 1024, 3 seeds) the qualitative ordering is unchanged—learned $\approx$ baseline $<$ ear $<$ oracle—though larger networks shrink the premium itself (0.945 vs. 1.008) by coordinating better from raw observations.

6.1.1. Training Dynamics: Three Stories Before the True One

We report, as a methodological caution, that this benchmark produced three successive spurious conclusions before the converged one, each supported by earnest-looking evidence. At 150 training cycles (of the ~ 400 needed), hand-crafted listeners appeared destructive (−1.6 for exclusivity) and the learned-ear conditions appeared to close 30% of a then-larger gap—an effect that evaporated (−3%) when seeds were doubled, because every condition was still climbing and seed-to-seed differences in learning stage masqueraded as treatment effects. Mid-training, value-of-information weighting of R_{comm} (paying for clarity before the scout’s pickup, when the supporter’s decision hangs) measurably steered emergence where uniform weighting bought only lazy clarity—decodability where behavior was already informative; at convergence both effects vanish. Communicative-MARL comparisons are unusually exposed to this hazard because the quantity of interest is a co-adaptation between policies and listeners, which converges more slowly than either alone. We recommend reporting the training-stage sensitivity of any claimed communication effect.

Table 2. Environment A at the converged budget (400 cycles; mean $\pm$ s.e.m.; $n=3$ seeds unless noted). “Gap” is the fraction of the oracle premium closed. Commit is the supporter’s commitment accuracy; R_{comm} on the information-gain scale.

Condition	R_{ext}	Gap	Commit	R_{comm}
Baseline ($n=10$)	$0.845 \pm .010$	—	.851	—
Oracle ($n=10$)	$0.957 \pm .008$	100%	.880	—
Simple L_0 ($n=10$)	$0.915 \pm .010$	62%	.857	-0.77
Exclusivity L_0	$0.863 \pm .010$	16%	.852	-1.77
Progress L_0 ($n=10$)	$0.924 \pm .008$	70%	.861	$+0.04$
Filter L_0 ($n=10$)	$0.920 \pm .006$	67%	.858	$+0.33$
Learned L_θ ($n=10$)	$0.828 \pm .015$	-15%	.853	$+1.03$
Learned + RSA	$0.850 \pm .010$	4%	.854	$+1.03$
Ear ($\lambda=0$, $n=6$)	$0.874 \pm .016$	26%	.854	—
Ear + R_{comm} ($n=10$)	$0.878 \pm .012$	30%	.858	$+1.04$
Filter + ear ($n=10$)	$0.920 \pm .012$	66%	.854	$+0.47$

6.2. Environment B: The Cost of Signaling with Nothing to Say

Table 3 summarizes final performance in the control environment; Figures 6–9 show training dynamics.

6.2.1. Hand-Crafted Listeners: Miscalibrated Signaling

The exclusivity listener exhibits an instructive failure mode, which we also observed in preliminary experiments with an independent, larger-scale implementation. The composite reward is clearly operative—early in training it even accelerates learning (Fig. 6), and it drives the geometric probe accuracy up faster than any other condition (Fig. 8)—yet the agent never comes close to optimizing the communicative reward itself. R_{comm} plateaus around -2.4 (Fig. 9), below the uninformative level of zero: the pragmatic listener built on this common ground assigns the true meaning less probability than a uniform guess. The exclusivity score (Eq. 7) demands that an action align with the true target’s direction while misaligning with every competing direction; when landmarks subtend similar bearings this margin is geometrically unsatisfiable, so the agent pays a permanent tax for a signal that cannot be produced. The tax is visible everywhere: final task reward drops by 47% relative to the baseline, distance and collision penalties rise, and specialization saturates lower (.815 vs. .847). The simple cosine listener, which removes the exclusivity margin, is correspondingly less harmful ($R_{\text{ext}} = 0.93$) and achieves a better—but still negative— R_{comm} . We call this failure mode miscalibrated signaling: a hand-crafted common ground rewards behavior the policy cannot realize and

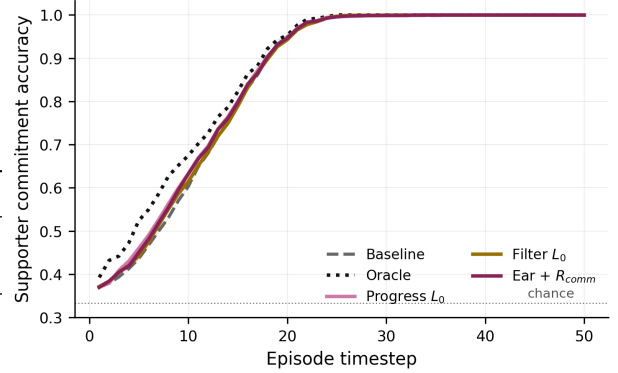


Fig. 1. Supporter commitment accuracy over the episode (converged policies, 256 held-out episodes). Motion-grounded listeners and the ear shift the curve toward the oracle’s during the decision window.

Table 3. Final performance (mean $\pm$ s.e.m. over 3 seeds, averaged over the last three evaluations). R_{ext} is the per-step team reward; Spec. is team specialization; R_{comm} is each condition’s own communicative reward (information-gain scale); Post-hoc is the decoding accuracy of a fresh listener trained on the final policy’s rollouts (chance = .33).

Condition	R_{ext}	Spec.	R_{comm}	Post-hoc
Baseline	$1.07 \pm .03$	.847	—	$.598 \pm .02$
Oracle	$0.99 \pm .04$	.841	—	$.589 \pm .01$
Simple L_0	$0.93 \pm .06$	.833	-1.33	$.599 \pm .01$
Exclusivity L_0	$0.57 \pm .04$	.815	-2.39	$.604 \pm .02$
Learned L_θ	$1.03 \pm .08$	.847	-0.13	$.582 \pm .03$
Learned + RSA	$1.09 \pm .03$	.853	-0.13	$.578 \pm .02$

an observer cannot decode, converting the intrinsic reward into pure interference.

6.2.2. Learned Listeners: Calibrated Legibility

The learned listener changes the picture qualitatively. Both learned conditions steadily optimize their communicative reward from -0.65 toward zero (Fig. 9), confirming that the co-adapted reward is learnable—by construction, it only credits distinctions that are actually present in the behavioral distribution. This calibration eliminates the destructive interference: task performance is statistically indistinguishable from the baseline ($1.03 \pm .08$ and $1.09 \pm .03$ vs. $1.07 \pm .03$), with the pragmatic variant nominally best on task reward and specialization. During training, the co-adapting listener’s accuracy climbs from chance ($1/3$) to $\approx .43$ and is still rising at the end (Fig. ??), with the RSA-recursed variant

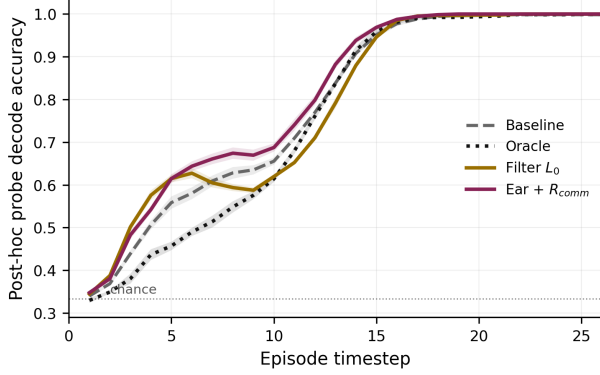


Fig. 2. Information profile: per-timestep decode accuracy of a fresh post-hoc probe on each condition’s converged behavior (first 26 steps shown; all saturate by $t \approx 16$). During the early decision window ($t \lesssim 10$) the filter-listener and ear-augmented scouts carry the most decodable intent, while oracle scouts—whose teams never rely on motion—are the least legible early. This is the mechanism behind the recovered gap: legibility is bought exactly when the partner is still deciding.

consistently ahead of the purely literal one—suggesting that contrastive reasoning over alternatives extracts signal that maximum-likelihood decoding alone does not.

6.2.3. How Much Was There to Say? Oracles and Post-Hoc Probes

Two further measurements put the above in perspective. First, the oracle: giving agents their teammates’ true preferences does not improve task performance (0.99 vs. 1.07). Near-optimal coordination in this environment is achievable from positions and velocities alone—who is closest to what resolves most conflicts—so the marginal task value of intent information is small. Second, the post-hoc probes: a fresh listener trained on any condition’s converged behavior decodes intentions at .58–.60 accuracy (Table 3), with no significant differences across conditions. Because acting on one’s preference is extrinsically rewarded, converged behavior is already informative; at $\lambda = 0.1$ there is simply little decodable information left for a calibrated reward to add—which the learned listener correctly reflects by saturating near zero, and which the hand-crafted listeners ignore by charging a permanent task tax for no informational return.

Together these results reframe the evaluation of implicit-communication methods: task reward alone conflates the cost of signaling with its value, and a method should be judged on (a) whether its intrinsic reward is achievable (calibration), (b) how much decodable information the resulting behavior carries (post-hoc

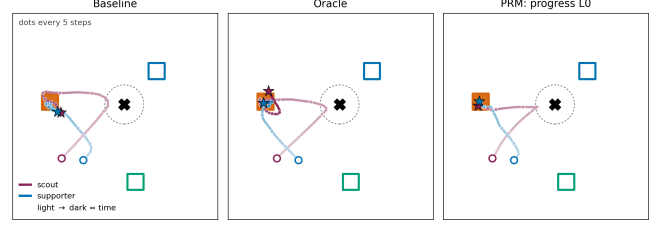


Fig. 3. A held-out Environment A episode illustrating the task. Squares are sites (the true target filled), the dashed circle is the pickup waypoint the scout must visit first. Trajectories darken with time (dot every 5 steps; dense dots = slow motion). The scout (dark red) detours through the waypoint before heading to the target; the slower supporter (blue) must infer the target from that motion. Open circle = start, star = end.

probes), and (c) what it pays for that information (task-reward gap). On all three criteria the learned listener dominates the hand-crafted ones.

6.2.4. The Weight λ : Dial or Landmine?

We sweep the communicative weight for both listener families in Environment B (2 seeds per new point; Fig. 4). With the learned listener, λ behaves like a dial along an information-performance frontier: $\lambda = 0.1$ is free ($R_{\text{ext}} = 1.03$, $R_{\text{comm}} = -0.13$); $\lambda = 0.3$ buys near-saturated clarity ($R_{\text{comm}} = -0.05$) for a modest task cost ($R_{\text{ext}} = 0.84$); and $\lambda = 1.0$ pushes the communicative reward to genuinely positive information gain ($R_{\text{comm}} = +0.62$)—behavior more informative than task-optimal motion—at a large task cost ($R_{\text{ext}} = -0.80$) as agents privilege signaling over covering landmarks. With the exclusivity listener, no operating point buys information: at $\lambda = 0.03$ the reward is simply ignored (R_{comm} stuck at -2.6 , task performance at baseline level), and at $\lambda = 0.1$ it destroys 47% of task reward with R_{comm} still at -2.4 . Calibration is precisely what turns λ from a landmine into a dial.

6.2.5. Qualitative Behavior

Figure 5 shows a representative held-out episode of Environment B. The baseline reaches landmarks but meanders, with late target switches. The exclusivity-listener agent displays the miscalibration diagnosed above: trajectories sweep across the arena and settle far from any landmark—costly motion signaling nothing. The learned-pragmatic agent produces direct, mutually distinct paths: each agent claims a separate landmark, and the agent whose preferred landmark is taken by a teammate cleanly re-targets the remaining one.

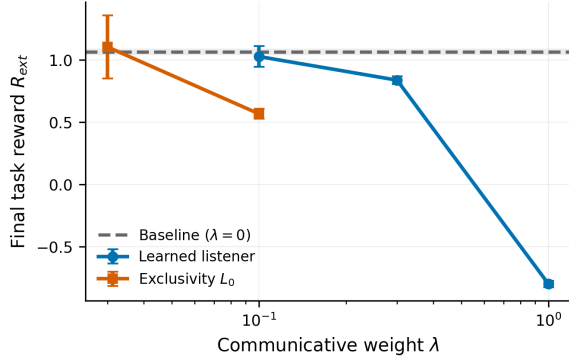


Fig. 4. Final task reward vs. communicative weight λ . The learned listener trades task reward for information smoothly; the exclusivity listener offers no λ at which information is gained.

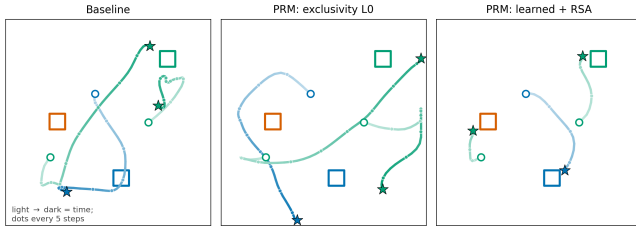


Fig. 5. A held-out episode under three conditions. Squares are landmarks (color = category); each trajectory is colored by the agent’s private preference and darkens with time, with a dot every 5 steps, so dense dots mean slow or stationary motion. Open circle = start, star = end.

7. DISCUSSION

The common ground is the problem. Our central empirical finding is that the success of a pragmatic reward hinges almost entirely on the alignment between the literal listener and the agents’ realizable behavior. A hand-crafted L_0 encodes the designer’s theory of what signaling should look like; when that theory is even mildly wrong—as with the exclusivity margin, which demands geometrically unsatisfiable action patterns—the recursion built on top of it faithfully amplifies the error, and the agent is paid to produce elaborate, undecodable gestures. This failure is quiet: behavioral metrics (path deviation, action magnitude, early probe accuracy) suggest the agent is “signaling harder,” while the communicative channel carries nothing. We suspect this failure mode is widespread in reward designs that score behavior against idealized observer models, and recommend a simple diagnostic that requires no extra machinery: track whether the agent can drive its own intrinsic reward above the

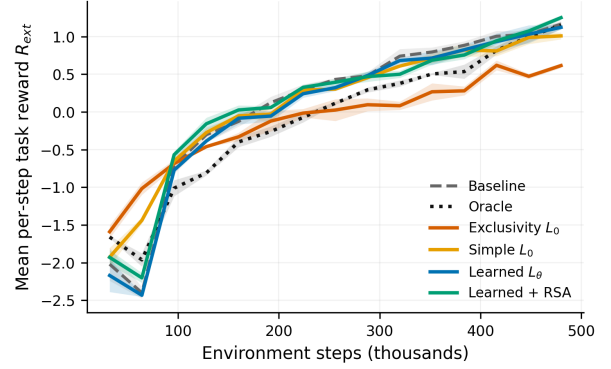


Fig. 6. Mean per-step team reward R_{ext} (3 seeds, mean $\pm$ s.e.m.).

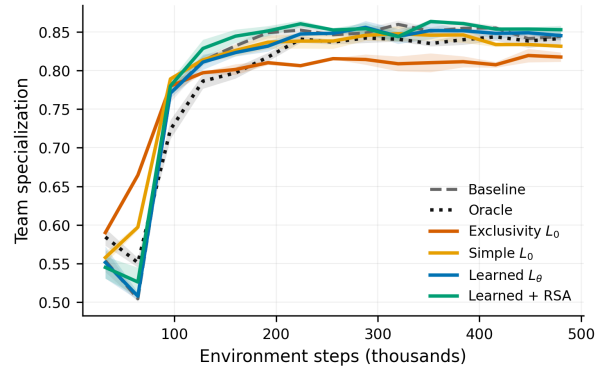


Fig. 7. Team specialization: fraction of landmarks claimed uniquely.

uninformative level. A persistently negative information gain is a certificate of miscalibration.

Learning the listener closes the loop. Training L_θ on the agents’ own replay data guarantees that the common ground never demands what the policy distribution does not contain; the reward then only sharpens distinctions that already exist, which is why it optimizes smoothly and costs no task performance. Proposition 1 explains the mechanism: the learned-listener reward is a variational mutual-information bound, so policy improvement against it is information maximization—grounded, by construction, in a decodable channel. And because the bound is calibrated, its weight becomes meaningful: λ moves the policy along an actual information-performance frontier (Sec. 6.2.4) instead of scaling an arbitrary shaping term. The RSA recursion contributes during training: it consistently improves the co-adapting listener’s accuracy, suggesting that pragmatic contrast (scoring an action against foregone alternatives) accelerates the emergence of decodable behavior, even where asymptotic decodability is capped by the task.

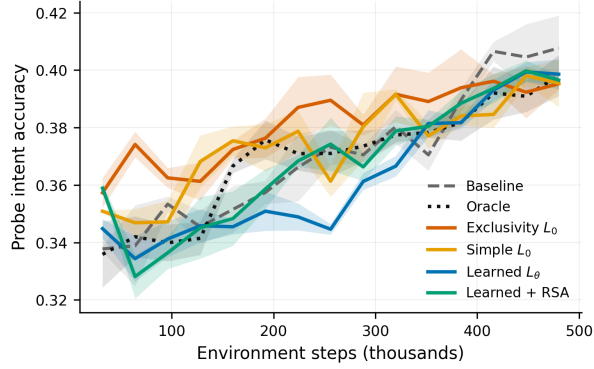


Fig. 8. Probe intent accuracy under the fixed geometric probe listener, identical across conditions.

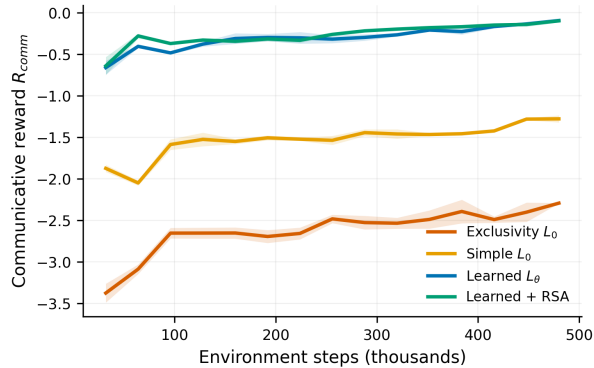


Fig. 9. Achieved communicative reward R_{comm} (each condition under its own listener; information-gain scale, 0 = uninformative).

Communication is a two-sided market—and the sides can substitute. Environment A decomposes implicit communication into speaker and receiver contributions, and the converged results show two distinct routes across the gap. The overt-signal route shapes the speaker with a bounded listener until its motion is readable by the teammate’s ordinary perception; here all the machinery is on the speaker side, and the receiver needs nothing but its own policy network. The equipped-receiver route leaves the speaker’s task-driven behavior untouched and invests in decoding, deploying the training listener at execution. Both are decentralized-executable; in our benchmark the overt-signal route recovers about twice as much of the oracle premium, suggesting that when the intended reader is weak—an RL policy, plausibly a human—moving the burden of clarity onto the speaker is the more effective allocation. The joint-exploration pathology of emergent communication [15] appears here only transiently, as a mid-training phenomenon.

The literal listener must be bounded—by design.

Our strongest finding is an inversion that computational pragmatics anticipated. The learned listener, trained to convergence, decodes the scout from raw observations and motion history at 97% regardless of how the scout behaves: its reward saturates, the pragmatic gradient vanishes, and the policy is left unshaped. The crude geometric listeners, by contrast, only credit overtly legible motion—and overt motion is exactly what a feed-forward teammate reading raw observations can also exploit. In RSA terms: the literal listener L_0 is not a defect to be learned away; its boundedness is what makes the speaker’s optimization informative. An omniscient audience needs no speech. Practically, this yields an audience-matching principle for legibility rewards: anchor R_{comm} in a listener whose competence approximates the weakest intended reader of the behavior—an RL teammate, or a human—not the best decoder trainable from the data.

Clarity must be paid when it is needed—during learning. A second, dynamical lesson: mid-training, rewards that concentrate clarity where the partner’s decisions hang (our value-of-information weighting) visibly steer emergence, while uniformly-integrated clarity is spent on behavior that is already informative. At convergence these weighting effects wash out, but the trajectory through which a team reaches its equilibrium is shaped by when clarity is subsidized—a distinction invisible to asymptotic evaluation alone.

When does legibility pay? The contrast between environments is a caution for the field: in Environment B, where geometry resolves coordination, even a perfect oracle gains nothing, and a well-calibrated legibility reward correctly approaches a no-op; in Environment A the same machinery faces a certified premium and recovers most of it. Benchmarks for communicative MARL should report their oracle gap—without it, task reward conflates the cost of signaling with its value. Where the gap is large, our results argue for evaluating methods on three separable axes: calibration of the intrinsic reward, information actually transmitted when the partner needs it (time-resolved probes), and the task price paid for it—all measured at, and reported across, converged training budgets.

7.1. Limitations and Future Work

Several limitations qualify our findings and chart future work. First, the intrinsic reward densifies the learning signal, which can aid learning independently of legibility; our commitment-timing and probe metrics mitigate but do not eliminate this confound. Second, the audience-matching principle is demonstrated for one audience class (feed-forward policies over raw observations); its quantitative form—how listener competence should scale

with reader competence—is open, and the wide-model check suggests the premium itself shrinks as readers strengthen. Third, meanings are hand-specified discrete variables; extending R_{comm} to continuous, hierarchical, or self-assigned meaning spaces—where the agent learns what is worth communicating, not only how—remains open. Fourth, validation is confined to AI-AI teams in 2D particle worlds; the audience-matching principle makes a testable prediction for human-agent teams—rewards anchored in human-competence listeners (e.g., L_θ fit to human intent judgments) should transfer better than either extreme—which we view as the natural next experiment. Finally, the framework is symmetric in an intriguing way: inverting the sign of λ turns legibility into obfuscation, suggesting privacy-preserving applications where agents remain opaque to opponents while staying legible to allies [4].

8. CONCLUSION

We presented a framework that unifies pragmatic inference and maximum-entropy reinforcement learning, importing the RSA listener value as a principled intrinsic reward for legible behavior, with the literal listener—the assumed common ground—as its decisive design choice, and learned listeners recovering information-theoretic approaches as a special case. Our two-environment study then locates where the choice bites. Where communication has no task value, miscalibrated listeners charge for undecodable signaling while calibrated ones correctly approach a no-op. Where communication is certifiably valuable, the decisive property is not listener competence but listener boundedness: crude motion-grounded listeners, which only credit overt signals, recover 60–70% of the oracle premium through the speaker-side reward alone, while a listener competent enough to decode anything demands nothing and shapes nothing—and is better spent as the teammate’s execution-time ear. This audience-matching principle is, we believe, RSA’s oldest lesson restated for reward design: speech is optimized against a literal listener precisely because the literal listener is limited. Agents whose competence includes being understood must be trained against audiences that need the help.

A. HYPERPARAMETERS

Table 4. Hyperparameters (all conditions unless noted).

Parameter	Value
Parallel environments	64
Episode length	50
Training cycles (Env. B / Env. A)	150 / 400
Actor / critic architecture	MLP 256×256
Learning rate (actor, critic)	3×10^{-4}
Discount γ	0.95
Target smoothing τ	0.005
Batch size / updates per env step	512 / 4
Entropy target	$-\dim(\mathcal{A})$ (auto-tuned)
Listener L_θ	MLP 128×128 , lr 10^{-3}
Direction listener temp. $\alpha_{\text{temp}} / K / N_r$	5 / 64 / 1
Learned+RSA alternatives / N_r	16 / 1
Filter listener temp. / forgetting ρ	2 / 0.9
λ (Env. B / Env. A)	0.1 / 0.3
VOI post-key weight (Env. A)	0.2
Scout / supporter max speed (Env. A)	1.0 / 0.6
Scout start annulus radius (Env. A)	[0.9, 1.3]
Bonus / cover / pickup radius (Env. A)	3.0 / 0.3 / 0.3
Motion-history window (Env. A)	6 steps
Seeds (headline / other conditions)	10 / 3

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